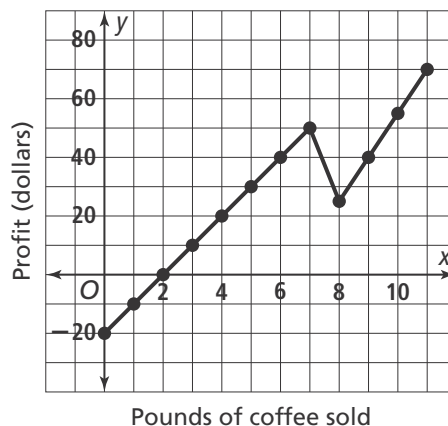


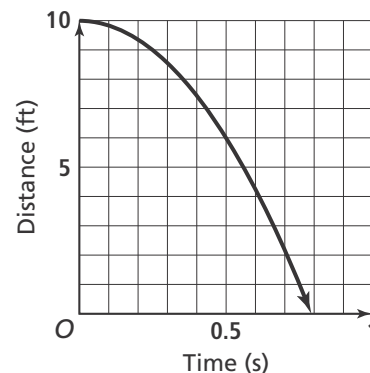
1-1 Lesson Quiz

Key Features of Functions

1. Benito produces and sells bulk coffee beans. The graph shows his profit as a function of the pounds of coffee beans sold. After selling a certain amount of coffee, Benito uses some earnings to buy more materials and then increases the price of the coffee. Which of the following is true?



- (A) Benito increases the price after selling 50 pounds of coffee beans.
- (B) The y-intercept represents the cost of one pound of coffee beans.
- (C) The x-intercept represents the number of pounds of coffee he has to sell before making a profit.
- (D) The domain of the function is $\{x \mid x \text{ is a real number greater than or equal to } 0\}$.
2. The graph of the function $f(x) = -16x^2 + 10$ shown at the right represents the distance, in feet, between a diver and the surface of the water, where x is the number of seconds after the diver leaves the diving board. What is the height of the diving board?



- feet
3. Which of the following correctly compares the two parts of the function in Item 1, over the intervals $[0, 7]$ and $[8, 11]$?
- (A) Benito made an extra \$25 after selling 7 pounds of coffee.
- (B) Benito spent \$1 after selling 50 pounds of coffee.
- (C) Benito's profit per pound of coffee decreases from \$50 to \$25.
- (D) Benito's profit per pound of coffee increases from \$10 to \$15.

Use the graph shown.

4. Identify the intervals on which the function is increasing and on which the function is decreasing.

Intervals on which the function increases: $(-\infty, \quad)$

Intervals on which the function decreases: $(\quad, \infty)$

5. At what point does the function switch from increasing to decreasing? $(\quad, \quad)$

